

Nithik R

Independent Researcher

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Education

National Institute of Technology Warangal

Aug 2025–Present

Bachelor of Science

- CGPA: 9.82/10

Academic interests: cryptography, cybersecurity, mathematics, quantum information, theoretical computer science.

Research / Writing

Cryptanalysis of a Quantum Public-Key Encryption Scheme

Aug 2026

- Conducted an independent cryptanalysis of the published quantum public-key encryption scheme “*Quantum public key encryption scheme with four states key*” (*Physica Scripta*, 97, 045102).
- Derived a simplified equivalent representation of the proposed complex encryption mechanism.
- Demonstrated that the construction **fails** to provide the **information-theoretic security** claimed by the original work.
- Formalised the analysis into a research comment and submitted the manuscript to *Physica Scripta*. Manuscript currently under peer review.
- Published the work to IACR Cryptology ePrint Archive, providing a publicly accessible research record.
- Preprint: [2026/1706](https://eprint.iacr.org/2026/1706)

Recu: Personal Website

Sep 2025–Present

- Maintains a personal technical website documenting work in **cryptography, cybersecurity, quantum information, and systems security**.
- Writes detailed technical articles and security analyses.
- Uses technical writing as a means of formalizing independently developed understanding and documenting growth.
- Website: recu.blog

Experience

BlitzHack

Remote

CTF Player

Oct 2025–Present

- Contributes to a national-level cybersecurity / CTF team focused on hands-on problem solving across **cryptography, binary exploitation, OSINT and forensics**.
- Participates in competitive security challenges and develop practical skills in vulnerability identification, exploit construction, and challenge creation.
- Placed **rank 5 in India** on CTFtime 2025.

Software Development Club

NIT Warangal

Cybersecurity Member

Jan 2026–Present

- Regularly conducts knowledge-sharing sessions which include talks, discussions and collaborative projects.
- Helps organise events, games, CTFs and awareness initiatives aimed at fostering interest in cybersecurity

ACM Summer School on Hardware Security

IIT Bombay

Participant

Jun 2026

- Selected to attend an intensive summer school focused on **hardware security and applied cybersecurity**.
- Studied security topics including **side-channel attacks, fault injection, microarchitectural attacks, post-quantum cryptography, and security of embedded / wireless systems**.
- Gained practical exposure to hardware-security methodology and open-source electronic-design-automation and formal-verification workflows, including **Icarus Verilog, Verilator, Yosys and GTKWave**.

Projects

fmevent: Semantic Event Conditions for the Shell (macOS CLI)

Oct 2026

- Built a command-line tool that lets shell scripts branch on the **meaning of text** rather than string matches: pipe text in, ask a plain-English question, and receive an exit status (0 yes, 1 no, 2 failure).
- Integrated Apple’s **on-device Foundation Model** through the fm CLI, so logs, diffs and source never leave the machine. Delivered as a single dependency-light POSIX **shell** script using jq.
- Designed a **semantic router** (priority ranking, composite routes derived with `all_of`, no-shell argv execution) and a **JSON-schema extraction** mode for structured output.
- Tested prompt-injection behaviour of a small on-device model and documented its limits. Added injection warnings, a `--strict` mode that requires model evidence to appear in the input, and `--greedy` sampling for reproducible CI runs.
- Wrote a **51-case offline test suite** using a fake model backend, covering exit codes, routing logic, timeouts and failure modes.
- GitHub: [noinoiexists/fmevent](https://github.com/noinoiexists/fmevent)

Post-Quantum Secure Pastebin

Jul 2026–Present

- Developing an open-source secure pastebin as a hands-on study of **modern cryptography, secure systems design and post-quantum security**.

- Using the project to study end-to-end concerns including confidentiality, integrity, key management, authenticated encryption, secure storage and protocol composition.

Binary Exploitation / Buffer Overflow Analysis

Apr 2026

- Studied stack memory organization and binary exploitation through the **OverTheWire Narnia** wargame.
- Analysed vulnerable C programs and developed a practical understanding of **buffer overflows, control-flow manipulation memory layout and privilege escalation**.
- Documented the attack methodology and underlying system behaviour in a technical write-up.
- Write-up: [Recu](#)

Data Encryption Standard Implementation

Aug 2025

- Implemented the **Data Encryption Standard (DES)** directly in C from the published specification, including the Feistel structure, permutations, key scheduling, S-box substitution, and round transformations.
- Implemented command-line encryption / decryption workflows and **ECB and CBC modes of operation**.
- Used the implementation to study block-cipher internals, mode-of-operation behaviour, and practical cryptographic engineering.
- GitHub: [noinoiexists/Data-Encryption-Standard](#)

Fibonacci LFSR Cipher and Known-Plaintext Cryptanalysis

Sep 2025

- Implemented a Fibonacci linear-feedback shift-register based cipher and investigated its underlying linear structure.
- Developed a **known-plaintext attack** exploiting the linear structure to recover the keystream / internal state and demonstrate the weakness of the construction.
- GitHub: [noinoiexists/Fibonacci-LFSR](#) | Write-up: [Recu](#)

Mathematical Algorithms

Sep 2025

- Implemented some algorithms in C, including **Gaussian elimination, Gauss–Jordan elimination, Gauss–Seidel iteration, LU decomposition and Gaussian elimination over GF(2)** for personal use.
- Focused on low-level implementation of mathematical algorithms relevant to cryptography and computational linear algebra.
- Conducted tests to estimate their efficiency.
- GitHub: [noinoiexists/Math-Tools](#)

Skills

Programming	C/C++, Python, JavaScript, TypeScript, HTML, CSS
Systems	GNU/Linux, Git, GitHub, Shell Scripting, CLI Tooling, Computer Architecture
Security	Cryptography, Cryptanalysis, Binary Exploitation, OSINT, Forensics
Mathematics	Linear Algebra, Number Theory, Abstract Algebra
Research & Technical Writing	Coherent research manuscripts and technical documentation

Updated: October 5, 2026